

1. Equality Check

Write a Python code to accept two integers and check whether they are equal or not.

Test Data : 15 15

Expected Output :

Number1 and Number2 are equal

```
In [5]: a=int(input())
b=int(input())
if a==b:
    print("Number1 and Number2 are equal")
else:
    print("Number1 and Number2 not equal")
```

Number1 and Number2 are equal

2. Even or Odd Check

Write a Python code to check whether a given number is even or odd.

Test Data : 15

Expected Output :

15 is an odd integer

```
In [6]: n=int(input())
if n%2==0:
    print(n,"is even number")
else:
    print(n,"is odd number")
```

15 is odd number

3. Positive or Negative Check

Write a Python code to check whether a given number is positive or negative.

Test Data : 15

Expected Output :

15 is a positive number

```
In [7]: n=int(input())
if n>0:
    print(n,"is a positive")
else:
    print(n,"is a negative")
```

15 is a positive

4. Voting Eligibility

Write a Python code to read the age of a candidate and determine whether he is eligible to cast his/her own vote.

Test Data : 21

Expected Output :

Congratulation! You are eligible for casting your vote.

```
In [8]: age=int(input())
if age>=18:
    print("Congratulation! You are eligible for casting your vote.")
else:
    print("Sorry! ,you are not eligible for casting your vote.")
```

Congratulation! You are eligible for casting your vote.

5. Largest of Three Numbers

Write a Python code to find the largest of three numbers.

Test Data : 12 25 52

Expected Output :

1st Number = 12

2nd Number = 25

3rd Number = 52

The 3rd Number is the greatest among three

```
In [9]: a=int(input())
b=int(input())
c=int(input())
if a>b and a>c:
    print("The 1st Number is the greatest among three")
if b>a and b>c:
    print("The 2nd Number is the greatest among three")
if c>a and c>b:
    print("The 3rd Number is the greatest among three")
```

The 3rd Number is the greatest among three

6. Coordinate Quadrant Identification

Write a Python code to accept a coordinate point in an XY coordinate system and determine in which quadrant the coordinate point lies.

Test Data : 7 9

Expected Output :

The coordinate point (7,9) lies in the First quadrant.

```
In [10]: x=int(input())
y=int(input())
if x>0 and y>0:
    print("The coordinate point",(x,y), "lies in the First quadrant.")
elif x<0 and y>0:
    print("The coordinate point",(x,y), "lies in the Second quadrant.")
elif x<0 and y<0:
    print("The coordinate point",(x,y), "lies in the Third quadrant.")
elif x>0 and y<0:
    print("The coordinate point",(x,y), "lies in the fourth quadrant.")
else:
    print("The coordinate point",(x,y), "lies at the origin .")
```

The coordinate point (7, 9) lies in the First quadrant.

7. Admission Eligibility Check

Write a Python code to determine eligibility for admission to a professional course based on the following criteria: Eligibility Criteria : Marks in Maths ≥ 65 and Marks in Phy ≥ 55 and Marks in Chem ≥ 50 and Total in all three subject ≥ 190 or Total in Maths and Physics ≥ 140 ----

Input the marks obtained in Physics :65

Input the marks obtained in Chemistry :51

Input the marks obtained in Mathematics :72

Total marks of Maths, Physics and Chemistry : 188

Total marks of Maths and Physics : 137

The candidate is not eligible.

Expected Output : The candidate is eligible for admission.

```
In [12]: py=int(input("Enter the marks obtained in the physics"))
ch=int(input("Enter the marks obtained in the chemistry"))
mt=int(input("Enter the marks obtained in the maths"))
total_marks=py+ch+mt
total_math_Py=mt+py
if mt>=65 and py>=55 and ch>=50:
    print(" The candidate is eligible for admission.")
elif total_math_Py>=140:
    print(" The candidate is eligible for admission.")
else:
    print(" The candidate is not eligible for admission.")
```

The candidate is eligible for admission.

8. Leap Year Determination

Write a Python code to find whether a given year is a leap year or not.

Test Data : 2016

Expected Output :

2016 is a leap year

```
In [4]: year=int(input())
if year%400==0:
    print(year,"is a leap year")
elif year%100==0:
    print(year,"is not leap year")
elif year%4==0:
    print(year,"is a leap year")
else:
    print(year,"is not leap year")
```

2000 is a leap year

9. Quadratic Equation Roots

Write a Python code to calculate the root of a quadratic equation.

Test Data : 1 5 7

Expected Output :

Root are imaginary;

No solution.

```
In [15]: a=float(input())
b=float(input())
c=float(input())

d=b**2-4*a*c
if d>0:
    root1=(-b+d**0.5)/(2*a)
    root2=(-b-d**0.5)/(2*a)
    print("Root1",root1,"Root2",root2)
elif d==0:
    root=-b/(2*a)
    print("both roots are equal:",root)
else:
    print("Root are imaginary")
    print("no solution")
```

Root are imaginary
no solution

10. Student Marks and Division Calculation

Write a Python code to read the roll no, name and marks of three subjects and calculate the total, percentage and division. Test Data :

Input the Roll Number of the student :784

Input the Name of the Student :James

Input the marks of Physics, Chemistry and Computer Application : 70 80 90

Expected Output :

Roll No : 784

Name of Student : James

Marks in Physics : 70

Marks in Chemistry : 80

Marks in Computer Application : 90

Total Marks = 240

Percentage = 80.00

Division = First

```
In [19]: rol_nu=input("enter the roo number")
name=input("Enter the name")
py=int(input("enter the physics's marks"))
ch=int(input("Enter the marks of chemistry"))
cs=int(input("Enter the marksof computer science"))
total_marks=py+ch+cs
per=total_marks/3
if per>60:
    div="First"
elif per>50:
    div="second"
elif per>40:
    div="third"
else:
    div="fail"
print("Roll no",rol_nu)
print("name",name)
print("total_marks",total_marks)
print("Percentage=",per)
print("Division",div)
```

Roll no 784

name james

total_marks 240

Percentage= 80.0

Division First

In []: